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An indirect multicriteria group decision-making method with heterogeneous preference relations and reliabilities of decision-makers

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ABSTRACT

This study proposes a new indirect method to solve multicriteria group decision-making (MCGDM) problems with heterogeneous preference relations and decision-makers' reliabilities. To unify the heterogeneous preference relations, the transformation of multiplicative preference relations (MPRs), fuzzy preference relations (FPRs), intuitionistic fuzzy preference relations (IFPRs), and linguistic preference relations (LPRs) into distributed preference relations (DPRs) is analyzed. As consistency among DPRs is constructed under the assumption that decision-makers are bounded rationality, the abstract transformation of MPR, FPR, IFPR, and LPR into DPR is defined, and its reality depends on the preferences of decision-makers. Two important properties of abstract transformation, namely, internal consistency and restricted max-max transitivity, were theoretically verified. The reliabilities of decision-makers are involved in MCGDM by estimating the reliability of each criterion to improve solution quality. Based on the four types of abstract transformations, a process for analyzing MCGDM problems with heterogeneous preference relations and decision-makers' reliabilities is developed. The proposed method is used to analyze a new business selection problem for an enterprise located in Changzhou, Jiangsu, China to demonstrate its applicability and validity.

1. Introduction

With the rapid development of emerging information technologies, such as big data, cloud computing, and the Internet of Things, socioeconomic activities have become more convenient and intelligent. A large amount of information related to socioeconomic activities was recorded under these conditions. Relevant information is useful for understanding socioeconomic activities, comprehensively and insightfully. Meanwhile, people are facing the challenge of solving real problems based on large amounts of information, which has increased exponentially [1]. Real problems involving large amounts of information are more complex. Individual intelligence is not sufficient to handle such problems because of limitations in knowledge and experience, and, thereby, group intelligence becomes an inevitable choice [2,3].

Many multicriteria group decision-making (MCGDM) methods have been developed (e.g., [4,5]). Usually, preference information

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of decision-makers can be characterized into two modes: comparing alternatives in pairs and evaluating alternatives individually. Each of the two modes is useful for MCGDM in different contexts. The first mode is the indirect evaluation of alternatives, whose representative preference styles include the fuzzy preference relation (FPR) [2,6], intuitionistic fuzzy preference relation (IFPR) [7,8], and distributed preference relation (DPR) [9]. The other mode is the direct evaluation of alternatives, whose representative preference styles include linguistic assessment [10], intuitionistic fuzzy numbers [11], and belief distribution [12]. This study focused on preference relations, characterizing comparisons between alternatives.

Owing to different cultural and educational backgrounds, knowledge, and experience, decision-makers are generally accustomed to selecting specific types of preference relations to characterize their preference information about a decision problem [13,14]. Representative types of preference relations include the multiplicative preference relation (MPR) [15,16], FPR [17,18], IFPR [19], linguistic preference relation (LPR) [20], and DPR [9]. MPR uses the scale of {1/9, …, 1/2, 1, …, 9} to characterize the ratio of preference intensity in experts' preferences of one alternative over another [15]. FPR uses an exact number from [0, 1] to characterize the preference degree in experts' preferences for one alternative over another [6]. IFPR uses the triple of $(\mu_{ij}, \nu_{ij}, \pi_{ij})$ to simultaneously characterize the preferred, non-preferred, and hesitant degrees in experts' preferences of one alternative over another [7]. LPR uses linguistic variables to characterize the preferred, non-preferred, or indifferent degrees of expert preference for one alternative over another [21]. DPR is defined as a distribution on a set of grades that can characterize the preferred, non-preferred, indifferent, and uncertain degrees of expert preference for one alternative over another simultaneously [9]. There is another type of preference relation called probabilistic language preference relation (PLPR) [22,23], whose form is very similar to that of DPR. Its differences from DPR are analyzed from the perspectives of uncertainty handling, consistency among preference relations, information aggregation, and elicitation process [24]. Considering these differences, the PLPR is not considered to be selected by decision-makers to characterize their preference information, which avoids the repetition of preference relations.

In addition to the different types of preference relations of decision-makers, the reliability of each decision-maker is another important perspective to consider in MCGDM [25]. The reliability of decision-makers is typically used to evaluate their proficiency in MCGDM [26]. To analyze MCGDM problems considering decision-makers' reliabilities, the reliability of each criterion [27] needs to be determined. Assuming that the criterion reliabilities are positively correlated with the criterion weights, they were specified by the facilitator in the study by Fu et al. [26]. When this assumption is relaxed, a heavy burden is placed on the facilitator to clearly differentiate the concepts of the weight and reliability of a criterion and specify them correctly. To relieve this burden, a reasonable and efficient estimation of criterion reliability has become an important issue.

To address MCGDM problems with heterogeneous preference relations and the consideration of decision-makers' reliabilities, different types of preference relations selected by different decision-makers must be handled first. There are three feasible methods for this purpose: indirect, direct, and optimization-based [28]. In the indirect method, different types of preference relations are transformed into a unified preference relation to generate solutions [29–30]. In the direct method, the individual priority vector of each decision-maker is derived from the decision-makers' preference relations and then aggregated as the collective priority vector to generate solutions [31]. In an optimization-based method, different types of preference relations are aggregated in an optimization model to determine the collective priority vector from which solutions are generated [32,33]. The priority vector can be derived from MPR, FPR, IFPR, and LPR, but not from DPR [1], which means that direct and optimization-based methods are not available for analyzing MCGDM problems in this study. The indirect method is the remaining option.

The development of an indirect method for analyzing MCGDM problems with MPRs, FPRs, IFPRs, LPRs, and DPRs as well as the consideration of decision-makers' reliabilities gives rise to three challenges. The first challenge is how to reflect the preferences of each decision-maker regarding the unification of the five types of preference relations. The second challenge is determining how unification can satisfy two important properties: namely internal consistency [34] and restricted max-max transitivity [35,36]. The third challenge is the reasonable and efficient estimation of the reliability of each criterion to ensure the rational involvement of decision-makers' reliabilities in MCGDM.

To address these challenges, this study proposes an indirect MCGDM method. To unify the five types of preference relations, the reasons why MPR, FPR, IFPR, and LPR are transformed into DPR are analyzed. The transformation from MPR, FPR, IFPR, and LPR into DPR is implemented based on abstract functions whose specific expressions are determined or selected by decision-makers to reflect their preference for the transformation. The first challenge is addressed in this study. For the transformation from MPR, FPR, IFPR, and LPR into DPR, based on abstract functions, the internal consistency and restricted max-max transitivity are theoretically proven to be satisfied. The second challenge is addressed in this study. The reliability of a criterion between a pair of alternatives for a decision-maker is estimated from the average similarity between the decision-maker's DPR and others' DPRs. The estimations of the reliability of the criterion for decision-makers are then combined by considering their divergence to be involved in the aggregation of DPRs for the pair of alternatives. The third challenge is addressed in this study. The proposed method was developed by transforming MPR, FPR, IFPR, and LPR into DPR based on abstract functions and estimating the reliability of each criterion.

The remainder of this paper is organized as follows. Section 2 introduces the modeling of MCGDM problems with MPRs, FPRs, IFPRs, LPRs, and DPRs. Section 3 analyzes why MPR, FPR, IFPR, and LPR are transformed into DPR. Section 4 describes the proposed method. In Section 5, the proposed method is used to analyze a new business selection problem for an enterprise located in Changzhou, Jiangsu, China. Finally, section 6 concludes the paper.

2. Modeling of MCGDM problems with five types of preference relations

Suppose there are M alternatives x_l ($l = 1, \ldots, M$) for an MCGDM problem, and each alternative is evaluated by T decision-makers t_j ($j = 1, \ldots, T$) based on L criteria e_i ($i = 1, \ldots, L$). The relative weights of L criteria are denoted by $w = (w_1, w_2, \ldots, w_L)$ such that $0 \le w_i \le 1$

and $\sum_{i=1}^{L} w_i = 1$. Considering the different expertise and experience of decision-makers, the facilitator specifies the relative weights of T decision-makers on criterion e_i ($i = 1, \ldots, L$), which are denoted by $\lambda(e_i) = (\lambda^1(e_i), \ldots, \lambda^T(e_i))$ such that $0 \leqslant \lambda^j(e_i) \leqslant 1$ ($j = 1, \ldots, T$) and $\sum_{j=1}^{T} \lambda^j(e_i) = 1$.

Five types of preference relations, namely MPR, FPR, IFPR, LPR, and DPR, are used to help decision-makers express their preference information about alternatives in pairs. According to the cultural and educational background, expertise, and experience of each decision-maker, they were encouraged to select any of the five types of preference relations to characterize their preference information. Without loss of generality, let us assume that decision-maker t_j ($j = 1, \ldots, q_1$) selects MPRs to provide preference information, decision-maker t_j ($j = q_1 + 1, \ldots, q_2$) selects FPRs to provide preference information, decision-maker t_j ($j = q_2 + 1, \ldots, q_3$) selects IFPRs to provide preference information, decision-maker t_j ($j = q_3 + 1, \ldots, q_4$) selects LPRs to provide preference information, and decision-maker t_j ($j = q_4 + 1, \ldots, T$) selects DPRs to provide preference information. The five types of preference relations are introduced below.

Definition 1. [15] *Given $X = \{x_l, l = 1, \ldots, M\}$, let us suppose that decision-maker t_j ($j = 1, \ldots, q_1$) gives the MPR matrix on $X \times X$ for criterion e_i ($i = 1, \ldots, L$), $A_i^j = (a_{i,lm}^j)_{M\times M}$ such that $a_{i,lm}^j \in \{1/9, \ldots, 1/2, 1, \ldots, 9\}$ and $a_{i,lm}^j \cdot a_{i,ml}^j = 1$ ($\forall\, l, m \in \{1, \ldots, M\}$), where $a_{i,lm}^j$ represents the ratio of preference intensity of alternative x_l to x_m for criterion e_i. Specifically, $a_{i,lm}^j = 1$ indicates the indifference between alternatives x_l and x_m for criterion e_i, $a_{i,lm}^j = 9$ indicates the absolute superiority of alternative x_l over x_m on criterion e_i, and $a_{i,lm}^j \in \{1/8, \ldots, 1/2, 2, \ldots, 8\}$ indicates the intermediate preference ratio of alternative x_l to x_m for criterion e_i.*

Definition 2. [6] *Given $X = \{x_l, l = 1, \ldots, M\}$, let us suppose that decision-maker t_j ($j = q_1 + 1, \ldots, q_2$) gives the FPR matrix on $X \times X$ for criterion e_i ($i = 1, \ldots, L$), $P_i^j = (p_{i,lm}^j)_{M\times M}$ such that $p_{i,lm}^j + p_{i,ml}^j = 1$ ($\forall\, l, m \in \{1, \ldots, M\}$), where $p_{i,lm}^j$ represents the preference degree of alternative x_l over x_m on criterion e_i. $p_{i,lm}^j = 0.5$, indicates the indifference between alternatives x_l and x_m on criterion e_i, $p_{i,lm}^j = 1$ indicates the absolute superiority of alternative x_l over x_m on criterion e_i, and $0.5 < p_{i,lm}^j < 1$ indicates the intermediate superiority of alternative x_l over x_m on criterion e_i.*

There was a mutual transformation between the FPRs and MPRs [37]. The transformation from MPR $a_{i,lm}^j$ to FPR $p_{i,lm}^j$ is represented by

$$p_{i,lm}^j = 0.5 \cdot \left(1 + log_9 a_{i,lm}^j\right),\ \ l, m \in \{1, \ldots, M\}, \tag{1}$$

which can be used to deduce the transformation from FPR $p_{i,lm}^j$ into MPR $a_{i,lm}^j$.

Definition 3. [7] *Given $X = \{x_l, l = 1, \ldots, M\}$, let us suppose that decision-maker t_j ($j = q_2 + 1, \ldots, q_3$) gives the IFPR matrix on $X \times X$ for criterion e_i ($i = 1, \ldots, L$), $R_i^j = (r_{i,lm}^j)_{M\times M}$ with $r_{i,lm}^j = (\mu_{i,lm}^j, \nu_{i,lm}^j, \pi_{i,lm}^j)$ such that $0 \leqslant \mu_{i,lm}^j + \nu_{i,lm}^j \leqslant 1$, $\pi_{i,lm}^j = 1 - \mu_{i,lm}^j - \nu_{i,lm}^j$, $\mu_{i,lm}^j = \nu_{i,ml}^j$, $\mu_{i,ll}^j = \nu_{i,ll}^j = 0.5$, and $\pi_{i,ll}^j = 0$ ($\forall\, l, m \in \{1, \ldots, M\}$), where $\mu_{i,lm}^j$, $\nu_{i,lm}^j$, and $\pi_{i,lm}^j$ indicate the preferred, non-preferred, and hesitancy (uncertain) degrees of alternative x_l over x_m on criterion e_i, respectively.*

Definition 4. [21] *Given $X = \{x_l, l = 1, \ldots, M\}$ and a subscript-symmetric additive linguistic term set $S = \{s_\alpha \mid \alpha = -\tau, \ldots, -1, 0, 1, \ldots, \tau\}$, let us suppose that decision-maker t_j ($j = q_3 + 1, \ldots, q_4$) gives the LPR matrix on $X \times X$ for criterion e_i ($i = 1, \ldots, L$), $L_i^j = (l_{i,lm}^j)_{M\times M}$ such that $l_{i,lm}^j \oplus l_{i,ml}^j = s_0$ ($\forall\, l, m \in \{1, \ldots, M\}$), where $l_{i,lm}^j$ represents the preference degree of alternative x_l over x_m on criterion e_i. In detail, $l_{i,lm}^j = s_0$ indicates the indifference between alternatives x_l and x_m on criterion e_i, $l_{i,lm}^j = s_\tau$ indicates the absolute superiority of alternative x_l over x_m on criterion e_i, and $s_0 < l_{i,lm}^j < s_\tau$ indicates the intermediate superiority of alternative x_l over x_m on criterion e_i.*

Definition 5. [9] *Given $X = \{x_l, l = 1, \ldots, M\}$, let us suppose that $\Omega = \{H_1, H_2, \ldots, H_N\}$ with an odd number N represents a set of symmetrical grades, where $H_{(N+1)/2}$ indicates an indifferent grade between alternatives x_l and x_m, $H_{(N+1)/2+1}, \ldots, H_N$ indicate the grades with increasing preferred intensity of alternative x_l over x_m, and $H_1, \ldots, H_{(N+1)/2\text{-}1}$ indicate the grades with decreasing non-preferred intensity of alternative x_l over x_m. Then, decision-maker t_j ($j = q_4 + 1, \ldots, T$) gives the DPR matrix on $X \times X$ for criterion e_i ($i = 1, \ldots, L$), $D_i^j = (d_{i,lm}^j)_{M\times M}$ with $d_{i,lm}^j = \{(H_n, d_{n,i,lm}^j), n = 1, \ldots, N; (\Omega, d_{\Omega,i,lm}^j)\}$, where $d_{n,i,lm}^j$ with $0 \le d_{n,i,lm}^j \le 1$, $d_{n,i,lm}^j = d_{N-n+1,i,ml}^j$, and $\sum_{n=1}^{N} d_{n,i,lm}^j \leqslant 1$ represents the belief degree assessed to grade H_n, and $d_{\Omega,i,lm}^j$ with $d_{\Omega,i,lm}^j = d_{\Omega,i,ml}^j = 1 - \sum_{n=1}^{N} d_{n,i,lm}^j$ represents the belief degree of global ignorance (uncertainty) between alternatives x_l and x_m on criterion e_i.*

Subsequently, based on the definitions of the five types of preference relations, their unification is discussed to generate a solution to the MCGDM problem.

3. Analysis of MCGDM problems with five types of preference relations

As this study aims to propose an indirect method for generating a solution to the MCGDM problem, the five types of preference relations should be unified. The key is to select a unified type of preference relation, as demonstrated in the following section.

(1) MPR and FPR

MPR and FPR can be transformed mutually [37]. Therefore, FPR is considered the focus for discussing the unified type of preference relation. As demonstrated by Definitions 2 and 3, an FPR indicates the preferred, indifferent, or non-preferred degree of one alternative over another, whereas an IFPR indicates the preferred, non-preferred, and uncertain degree of one alternative over another. The preference information characterized by an IFPR cannot be equivalently described by an FPR. The transformation from IFPR to FPR inevitably results in information loss. As a result, MPR and FPR cannot be considered a unified type of preference relation.

(2) IFPR

As demonstrated by Definitions 3 and 5, an IFPR indicates the preferred, non-preferred, and uncertain degrees of one alternative over another, whereas a DPR indicates the preferred, non-preferred, indifferent, and uncertain degrees between alternatives for a set of symmetrical grades. Preference information characterized by preferred, non-preferred, indifferent, and uncertain grades cannot be described as a triple of the preferred, non-preferred, and uncertain degrees. The transformation from DPR to IFPR results in information loss. As a result, IFPR cannot be considered a unified type of preference relation.

(3) LPR

As demonstrated by Definitions 4 and 5, an LPR uses one linguistic variable to indicate the preferred, indifferent, or non-preferred degree of one alternative over another on a subscript-symmetric additive linguistic term set, whereas a DPR simultaneously indicates the preferred, non-preferred, indifferent, and uncertain degree between alternatives on a set of symmetrical grades. Preference information characterized by preferred, non-preferred, indifferent, and uncertain grades, cannot be described using only one linguistic variable. Converting DPR into LPR results in information loss. As a result, LPR cannot be considered a unified type of preference relation.

Based on the above analysis, MPR, FPR, IFPR, and LPR cannot be used to unify the five types of preference relations. The remaining option is DPR. Let us suppose that the MPRs, FPRs, IFPRs, and LPRs are unified as DPRs. As each decision-maker selects MPR, FPR, IFPR, LPR, or DPR to characterize the preference information, their preference should be reflected in the transformation from MPR, FPR, IFPR, or LPR into DPR. A detailed transformation is presented in Section 4.

After the unification of the decision-makers' preference information, the resulting DPRs are used to generate a solution to the MCGDM problem. In contrast to LPR, the score values of grades $s(H_n)$ ($n = 1, \ldots, N$) are used to characterize the difference among the grades, which satisfy $0 < s(H_{(N+1)/2+1}) < \ldots < s(H_n) = 1$, $s(H_{(N+1)/2}) = 0$, and $s(H_n) + s(H_{N\text{-}n+1}) = 0$ ($n \in \{1, \ldots, N\}$). To facilitate generating a solution with DPRs, the group DPR $D_{lm} = \{(H_n, d_n(a_{lm})), n = 1, \ldots, N; (\Omega, d_\Omega(a_{lm}))\}$ is combined with $s(H_n)$ ($n = 1, \ldots, N$) to generate a score interval $s_{lm} = [s_{lm}^-, s_{lm}^+]$ such that

$$s_{lm}^- = \sum_{n=1}^{N} d_n(a_{lm}) \cdot s(H_n) \ + \ d_\Omega(a_{lm}) \cdot s(H_1), \tag{2}$$

$$s_{lm}^+ = \sum_{n=1}^{N} d_n(a_{lm}) \cdot s(H_n) \ + \ d_\Omega(a_{lm}) \cdot s(H_N), \tag{3}$$

$$s_{ml}^- = -s_{lm}^+, \text{ and} \tag{4}$$

$$s_{ml}^+ = -s_{lm}^-. \tag{5}$$

Eqs. (2)-(3) indicate that $s_{lm} \subseteq [-1, 1]$. Score intervals s_{lm} and s_{ml} must be compared to determine whether alternative x_l is superior to x_m. To facilitate such a comparison, the score interval s_{lm} is transformed into a score S_{lm} based on its mean μ_{lm} and standard variance σ_{lm} using the score function of Fu et al. [38]:

$$S_{lm} = \mu_{lm} \cdot \left(1 - \frac{\sigma_{lm}}{\sigma_{\max}}\right) = \frac{s_{lm}^+ + s_{lm}^-}{2} \cdot \left(1 - \frac{s_{lm}^+ - s_{lm}^-}{\sqrt{12}} \bigg/ \frac{2}{\sqrt{12}}\right) = \frac{s_{lm}^+ + s_{lm}^-}{2} \cdot \left(1 - \frac{s_{lm}^+ - s_{lm}^-}{2}\right), \tag{6}$$

where $\sigma_{\max} = \frac{2}{\sqrt{12}}$.

The properties of S_{lm} are described as follows.

Property 1. *Given the score intervals $s_{lm} = [s_{lm}^-, s_{lm}^+]$ and $s_{ml} = [s_{ml}^-, s_{ml}^+]$, their scores S_{lm} and S_{ml}, calculated using Eq. (6) satisfy:*

(1) $S_{i,lm}^{j,\mu} > S_{i,ml}^{j,\mu}$ if $\mu_{i,lm}^j - \nu_{i,lm}^j > \mu_{i,ml}^j - \nu_{i,ml}^j$, and.

(2) $-1 \le S_{lm}, S_{ml} \le 1$.

The conclusions presented for Property 1 are clear from Eq. (6); therefore, the proof is omitted here. If $S_{lm} > S_{ml}$, the alternative x_l is said to be superior to x_m, and vice versa. Specifically, $S_{lm} = S_{ml}$ indicates that alternative x_l is indifferent to x_m.

4. Proposed method

Sections 2 and 3 describe the proposed method. The transformation from MPR, FPR, IFPR, and LPR into DPR is constructed using abstract functions to reflect the preferences of decision -makers regarding the transformation. Two important properties, namely internal consistency [34] and restricted max-max transitivity [35,36], have been theoretically proven to be satisfactory for the constructed transformation. The reliability of each criterion was estimated reasonably and efficiently. The process of generating a solution to the MCGDM problem is presented using the constructed transformation and estimation.

4.1. Transformation from MPRs and FPRs into DPRs

FPR and MPR can be converted into one another [37]. Considering this, the focus is on how to transform an FPR into a DPR. Such a transformation depends on constructing a mapping from the preference degree $p^j_{i,lm}$, such that $p^j_{i,lm}\in[0, 1]$, to the score value of grade H_n, $s(H_n)$, such that $s(H_n)\in[-1, 1]$. An abstract function is adopted to implement the mapping, in which the preference of a decision-maker can be reflected.

Definition 6. *Suppose, an FPR matrix $P^j_i = (p^j_{i,lm})_{M\times M}$ is presented in Definition 2, a DPR matrix $D^j_i = (d^j_{i,lm})_{M\times M}$ with $d^j_{i,lm} = \{(H_n, d^j_{n,i,lm}), n = 1, \ldots, N; (\Omega, d^j_{\Omega,i,lm})\}$ is presented in Definition 5, and the score values of grades $s(H_n)$ ($n = 1, \ldots, N$) are given. Then, the abstract function $f_1(x)$: $[0, 1] \to [-1, 1]$ is adopted perform mapping from the preference degree $p^j_{i,lm}$ to the score value of grade H_n, $s(H_n)$, where $f_1(x)$ satisfies the following condition*:

$$(1)\ \text{Continuity}: \quad \lim_{x\to x_0} f_1(x) = f_1(x_0), \forall x_0 \in [0, 1], \tag{7}$$

$$(2)\ \text{Monotonicity}: \quad f_1(x_1) > f_1(x_2) \text{ if } x_1 > x_2, \forall x_1, x_2 \in [0, 1], \tag{8}$$

$$(3)\ \text{Boundedness}: \quad f_1(0) = -1 \quad \text{and} \quad f_1(1) = 1, \quad \text{and} \tag{9}$$

$$(4)\ \text{Median}: \quad f_1(0.5) = 0. \tag{10}$$

Any specific function satisfying the four conditions presented in Definition 6 can be adopted by a decision-maker to help transform FPR into DPR. The adoption of a specific function depends on the preferences of the decision-maker. The transformation from FPR to DPR is defined using the abstract function described in Definition 6.

Definition 7. *Under the conditions described in Definition 6, an abstract function satisfying the four conditions in Eqs. (7)–(10), the preference degree $p^j_{i,lm}$ is transformed into DPR $d^j_{i,lm} = \{(H_n, d^j_{n,i,lm}), n = 1, \ldots, N\}$, with.*

$$d^j_{n,i,lm} = \begin{cases} \dfrac{f_1\left(p^j_{i,lm}\right) - s(H_{n-1})}{s(H_n) - s(H_{n-1})}, & s(H_{n-1}) \le f_1\left(p^j_{i,lm}\right) \langle s(H_n) \\ \dfrac{s(H_{n+1}) - f_1\left(p^j_{i,lm}\right)}{s(H_{n+1}) - s(H_n)}, & s(H_n) \le f_1\left(p^j_{i,lm}\right) \langle s(H_{n+1}) \\ 0, & \text{others} \end{cases}. \tag{11}$$

A numerical example is provided to illustrate the transformation of an FPR into a DPR using Eq. (11).

Example 1. *Suppose, $\Omega = \{H_1, H_2, \ldots, H_9\}$ and $s(H_n)$ ($n = 1, \ldots, 9$) = (-1, −0.75, −0.5, −0.25, 0, 0.25, 0.5, 0.75, 1). Given the specific function $f_1(x) = 2x–1$ and an FPR $p^j_{i,lm} = 0.3$, the transformation process from an FPR to a DPR is as follows. According to Eq. (11), we have.*

$f_1\left(p^j_{i,lm}\right)$ = 2·0.3 – 1 = -0.4,

$d^j_{3,i,lm} = \frac{s(H_{3+1}) - f_1\left(p^j_{i,lm}\right)}{s(H_{3+1}) - s(H_3)}$ = (-0.25 – (-0.4)) / (-0.25 – (-0.5)) = 0.6, and.

$d^j_{4,i,lm} = \frac{f_1\left(p^j_{i,lm}\right) - s(H_{4-1})}{s(H_4) - s(H_{4-1})}$ = (-0.4 – (-0.5)) / (-0.25 – (-0.5)) = 0.4.

Overall, $d^j_{i,lm} = \{(H_3, 0.6), (H_4, 0.4)\}$.

Example 1 shows the detailed transformation process from FPR to DPR. Using Definition 7, FPR matrices P^j_i ($i = 1, \ldots, L$, $j = q_1 + 1, \ldots, q_2$) can be transformed into DPR matrices D^j_i. The rationality of the transformation is strongly associated with two important properties: internal consistency [34] and restricted max-max transitivity [35,36].

Property 2. [34] *The rankings of the alternatives generated from the transformed preference relation matrix are the same as those from the original preference relation matrix.*

Property 3. [35,36] *Given three alternatives x_l, x_m and x_n, if alternative x_l is better than x_m and x_m is better than x_n, then the preference intensity of x_l over x_n is larger than and equal to the maximum one in the preference intensities of x_l over x_m and x_m over x_n.*

These two properties are demonstrated in the following.

Proposition 1. *Suppose, a DPR matrix $D^j_i = (d^j_{i,lm})_{M\times M}$ is transformed from an FPR matrix $P^j_i = (p^j_{i,lm})_{M\times M}$ by using Definitions 6 and 7. Under the conditions, the score of $d^j_{i,lm}$ is calculated as $S^{j,F}_{i,lm}$ using Eqs. (2)-(6). Then, the internal consistency between P^j_i and D^j_i is said to be satisfied when.*

(1) $S_{i,lm}^{j,F} > S_{i,ml}^{j,F}$ if $p_{i,lm}^{j} > 0.5$,
(2) $S_{i,lm}^{j,F} = S_{i,ml}^{j,F}$ if $p_{i,lm}^{j} = 0.5$, and
(3) $S_{i,lm}^{j,F} < S_{i,ml}^{j,F}$ if $p_{i,lm}^{j} < 0.5$.

Proposition 2. *Suppose, a DPR matrix $D_i^j = (d_{i,lm}^j)_{M\times M}$ is transformed from an FPR matrix $P_i^j = (p_{i,lm}^j)_{M\times M}$ by using Definitions 6 and 7. Assume that the scores of $d_{i,lm}^j$, $d_{i,mn}^j$ and $d_{i,ln}^j$, i.e., $S_{i,lm}^{j,F}$, $S_{i,mn}^{j,F}$ and $S_{i,ln}^{j,F}$ are obtained using Eqs. (2)-(6). Under the conditions, the restricted max-max transitivity of D_i^j is said to be satisfied when.*

(1) $S_{i,ln}^{j,F} \geq \max\{S_{i,lm}^{j,F}, S_{i,mn}^{j,F}\}$ if $p_{i,lm}^{j} \geq 0.5$, $p_{i,mn}^{j} \geq 0.5$, and $p_{i,ln}^{j} \geq \max\{p_{i,lm}^{j}, p_{i,mn}^{j}\}$, and.
(2) $S_{i,ln}^{j,F} \leq \min\{S_{i,lm}^{j,F}, S_{i,mn}^{j,F}\}$ if $p_{i,lm}^{j} \leq 0.5$, $p_{i,mn}^{j} \leq 0.5$, and $p_{i,ln}^{j} \leq \min\{p_{i,lm}^{j}, p_{i,mn}^{j}\}$.

Propositions 1 and 2 are proven in Sections A.1 and A.2 of Appendix A in the supplementary material. The above analysis verifies the rationality of the transformation from FPR to DPR, as presented in Definitions 6 and 7.

4.2. Transformation from IFPRs into DPRs

As presented in Definition 3, in IFPR $r_{i,lm}^j = (\mu_{i,lm}^j, \nu_{i,lm}^j, \pi_{i,lm}^j)$, $\pi_{i,lm}^j$ can be obtained from $\mu_{i,lm}^j$ and $\nu_{i,lm}^j$. Considering this fact, the uncertain degree $\pi_{i,lm}^j$ of IFPR can be transformed into the global ignorance $d_{\Omega,i,lm}^j$ of DPR, and the transformation from IFPR into DPR is dependent on constructing a binary mapping from $\mu_{i,lm}^j$ and $\nu_{i,lm}^j$ to the score value of grade H_n, $s(H_n)$ such that $s(H_n)\in[-1, 1]$. To reflect the preference of the decision-maker in mapping, an abstract function was adopted.

Definition 8. *Suppose, an IFPR matrix $R_i^j = (r_{i,lm}^j)_{M\times M}$ with $r_{i,lm}^j = (\mu_{i,lm}^j, \nu_{i,lm}^j, \pi_{i,lm}^j)$ presented in Definition 3, a DPR matrix $D_i^j = (d_{i,lm}^j)_{M\times M}$ with $d_{i,lm}^j = \{(H_n, d_{n,i,lm}^j), n = 1, \ldots, N; (\Omega, d_{\Omega,i,lm}^j)\}$ presented in Definition 5, and the score values of grades $s(H_n)$ ($n = 1, \ldots, N$) are given. Then, the abstract function $f_2(x, y)$: $[0, 1] \times [0, 1] \to [-1, 1]$ is adopted to carry out the mapping from the preferred and non-preferred degrees $\mu_{i,lm}^j$ and $\nu_{i,lm}^j$ to the score value of grade H_n, $s(H_n)$, where $f_2(x, y)$ satisfies the following condition*:

$$(1)\ \text{Continuity}:\quad \lim_{x\to x_0, y\to y_0} f_2(x,y) = f_2(x_0,y_0), \forall 0\leqslant x_0+y_0\leqslant 1, \tag{12}$$

$$(2)\ \text{Monotonicity}:\quad \begin{aligned} & f_2(x_1,y) > f_2(x_2,y) \text{ if } 1\geqslant x_1 > x_2\geqslant 0 \quad \text{and} \\ & f_2(x,y_1) > f_2(x,y_2) \text{ if } 0\leqslant y_1 < y_2\leqslant 1, \end{aligned} \tag{13}$$

$$(3)\ \text{Boundedness}:\quad f_2(0,1-\pi_{i,lm}^j) = -1 \quad \text{and} \quad f_2(1-\pi_{i,lm}^j,0) = 1, \quad \text{and} \tag{14}$$

$$(4)\ \text{Median}:\quad f_2(0.5-0.5\pi_{i,lm}^j, 0.5-0.5\pi_{i,lm}^j) = 0. \tag{15}$$

The decision-maker can select a specific function that satisfies the four conditions presented in Eqs. (12)–(15) to transform an IFPR to a DPR. The choice of a specific function reflects the decision-maker's preference. The transformation from IFPR to DPR is defined using the abstract function described in Definition 8.

Definition 9. *Under the conditions described in Definition 8, an abstract function satisfying the four conditions in Eqs. (12)–(15), the IFPR $r_{i,lm}^j$ is transformed into a DPR $d_{i,lm}^j = \{(H_n, d_{n,i,lm}^j), n = 1, \ldots, N; (\Omega, \pi_{i,lm}^j)\}$, with.*

$$d_{n,i,lm}^j = \begin{cases} \dfrac{f_2\left(\mu_{i,lm}^j, \nu_{i,lm}^j\right) - s(H_{n-1})}{s(H_n) - s(H_{n-1})}\cdot\left(\mu_{i,lm}^j + \nu_{i,lm}^j\right), & s(H_{n-1})\leqslant f_2\left(\mu_{i,lm}^j, \nu_{i,lm}^j\right) \langle s(H_n) \\ \dfrac{s(H_{n+1}) - f_2\left(\mu_{i,lm}^j, \nu_{i,lm}^j\right)}{s(H_{n+1}) - s(H_n)}\cdot\left(\mu_{i,lm}^j + \nu_{i,lm}^j\right), & s(H_n)\leqslant f_2\left(\mu_{i,lm}^j, \nu_{i,lm}^j\right) \langle s(H_{n+1}). \\ 0, & \text{others} \end{cases} \tag{16}$$

Using Definition 9, the IFPR matrices R_i^j ($i = 1, \ldots, L, j = q_2+1, \ldots, q_3$) can be transformed into the DPR matrices D_i^j. Properties 2 and 3 guarantee the rationality of the transformation from IFPR to DPR, as presented in Definitions 8 and 9. To examine the internal consistency between R_i^j and D_i^j, the two IFPRs $r_{i,lm}^j$ and $r_{i,ml}^j$ were compared. Considering that an IFPR $r_{i,lm}^j$ can be equivalently transformed to an interval number, $[\mu_{i,lm}^j, 1 - \nu_{i,lm}^j] \subseteq [0, 1]$ [39], the score of $[\mu_{i,lm}^j, 1 - \nu_{i,lm}^j]$ was calculated using Eq. (6) as

$$S_{i,lm}^{j,\mu} = \mu_{lm} \cdot \left(1 - \frac{\sigma_{lm}}{\sigma_{\max}}\right) = \frac{1 - \nu_{i,lm}^{j} + \mu_{i,lm}^{j}}{2} \cdot \left(1 - \frac{1 - \nu_{i,lm}^{j} - \mu_{i,lm}^{j}}{\sqrt{12}} \Big/ \frac{1}{\sqrt{12}}\right) = \frac{1 - \nu_{i,lm}^{j} + \mu_{i,lm}^{j}}{2} \cdot \left(\nu_{i,lm}^{j} + \mu_{i,lm}^{j}\right), \tag{17}$$

where $\sigma_{\max} = \frac{1}{\sqrt{12}}$.

The properties of $S_{i,lm}^{j,\mu}$ are described as follows.

Property 4. *Given the IFPRs $r_{i,lm}^{j} = [\mu_{i,lm}^{j}, 1 - \nu_{i,lm}^{j}]$ and $r_{i,ml}^{j} = [\mu_{i,ml}^{j}, 1 - \nu_{i,ml}^{j}]$, their scores $S_{i,lm}^{j,\mu}$ and $S_{i,ml}^{j,\mu}$ calculated using Eq. (17) satisfy:*

(1) $S_{i,lm}^{j,\mu} > S_{i,ml}^{j,\mu}$ if $\mu_{i,lm}^{j} - \nu_{i,lm}^{j} > \mu_{i,ml}^{j} - \nu_{i,ml}^{j}$, and
(2) $0 \le S_{i,lm}^{j,\mu}, S_{i,ml}^{j,\mu} \le 1$.

The conclusions presented in Property 4 are clear to hold from Eq. (17) and thus their proof is omitted here. With Eq. (17), the internal consistency between R_i^j and D_i^j is focused.

Proposition 3. *Suppose, a DPR matrix $D_i^j = (d_{i,lm}^{j})_{M\times M}$ is transformed from an IFPR matrix $R_i^j = (r_{i,lm}^{j})_{M\times M}$ by using Definitions 8 and 9. Under the conditions, the score of IFPR $S_{i,lm}^{j,\mu}$ and the score of DPR $S_{i,ml}^{j,I}$ are calculated using Eq. (17) and Eqs. (2)-(6), respectively. Then, the internal consistency between R_i^j and D_i^j is said to be satisfied when.*

(1) $S_{i,lm}^{j,I} \ge S_{i,ml}^{j,I}$ if $S_{i,lm}^{j,\mu} \ge S_{i,ml}^{j,\mu}$, and
(2) $S_{i,lm}^{j,I} < S_{i,ml}^{j,I}$ if $S_{i,lm}^{j,\mu} < S_{i,ml}^{j,\mu}$.

This proposition is proven in Section A.3 of Appendix A. As for the restricted max-max transitivity of IFPR matrix R_i^j, an assumption is given.

Assumption 1. *For the IFPR matrix R_i^j described in Definition 3, after decision-maker t_j has provided $r_{i,lm}^{j}$ and $r_{i,mn}^{j}$, the hesitant (uncertain) degree of $r_{i,ln}^{j}$ is reduced.*

Assumption 1 is based on the idea that decision-maker t_j has a clearer understanding of alternatives x_l, x_m and x_n after t_j has given $r_{i,lm}^{j}$ and $r_{i,mn}^{j}$. This assumption indicates that $u_{i,ln}^{j} + \nu_{i,ln}^{j} \ge \max\{u_{i,lm}^{j} + \nu_{i,lm}^{j}, u_{i,mn}^{j} + \nu_{i,mn}^{j}\}$. Under Assumption 1, the restricted max-max transitivity of D_i^j is examined.

Proposition 4. *Supposed, a DPR matrix $D_i^j = (d_{i,lm}^{j})_{M\times M}$ is transformed from an IFPR matrix $R_i^j = (r_{i,lm}^{j})_{M\times M}$ by using Definitions 8 and 9. Assume that the scores of $r_{i,lm}^{j}$, $r_{i,mn}^{j}$, $r_{i,ln}^{j}$, $d_{i,lm}^{j}$, $d_{i,mn}^{j}$ and $d_{i,ln}^{j}$, i.e., $S_{i,lm}^{j,\mu}$, $S_{i,mn}^{j,\mu}$, $S_{i,ln}^{j,\mu}$, $S_{i,lm}^{j,I}$, $S_{i,mn}^{j,I}$ and $S_{i,ln}^{j,I}$ are calculated using Eq. (17) and Eqs. (2)-(6). Under the conditions, the restricted max-max transitivity of D_i^j is said to be satisfied when.*

(1) $S_{i,ln}^{j,I} \ge \max\{S_{i,lm}^{j,I}, S_{i,mn}^{j,I}\}$ if $S_{i,lm}^{j,\mu} \ge S_{i,ml}^{j,\mu}$, $S_{i,mn}^{j,\mu} \ge S_{i,nm}^{j,\mu}$ and $S_{i,ln}^{j,\mu} \ge \max\{S_{i,lm}^{j,\mu}, S_{i,mn}^{j,\mu}\}$, and
(2) $S_{i,ln}^{j,I} \le \min\{S_{i,lm}^{j,I}, S_{i,mn}^{j,I}\}$ if $S_{i,lm}^{j,\mu} \le S_{i,ml}^{j,\mu}$, $S_{i,mn}^{j,\mu} \le S_{i,nm}^{j,\mu}$ and $S_{i,ln}^{j,\mu} \le \min\{S_{i,lm}^{j,\mu}, S_{i,mn}^{j,\mu}\}$.

We prove Proposition 4 in Section A.4 of Appendix A. This analysis verifies the rationality of the transformation from IFPR to DPR, as presented in Definitions 8 and 9.

4.3. Transformation from LPRs into DPRs

The transformation from LPR to DPR depends on constructing a mapping from the subscript α of $l_{i,lm}^{j}$, such that $\alpha \in [-\tau, \tau]$, to the score value of grade H_n, $s(H_n)$, such that $s(H_n) \in [-1, 1]$. An abstract function was adopted to implement the mapping. The selection of a specific function by a decision-maker reflects his or her preference.

Definition 10. *Suppose, an LPR matrix $L_i^j = (l_{i,lm}^{j})_{M\times M}$ presented in Definition 4, a DPR matrix $D_i^j = (d_{i,lm}^{j})_{M\times M}$ with $d_{i,lm}^{j} = \{(H_n, d_{n,i,lm}^{j}), n = 1, \ldots, N; (\Omega, d_{\Omega,i,lm}^{j})\}$ presented in Definition 5, and the score values of grades $s(H_n)$ ($n = 1, \ldots, N$). The abstract function $f_3(x)$: $[-\tau, \tau] \to [-1, 1]$ is then adopted to implement the mapping from the subscript α of $l_{i,lm}^{j}$ to the score value of grade H_n, $s(H_n)$, where $f_3(x)$ satisfies the following condition:*

$$(1)\ \text{Continuity}:\quad \lim_{x\to x_0} f_3(x) = f_3(x_0), \forall x_0 \in [-\tau, \tau], \tag{18}$$

$$(2)\ \text{Monotonicity}: f_3(x_1) > f_3(x_2),\ \text{if}\ x_1 > x_2, \forall x_1, x_2 \in [-\tau, \tau], \tag{19}$$

$$(3)\ \text{Boundedness}:\quad f_3(-\tau) = -1 \quad \text{and} \quad f_3(\tau) = 1, \quad \text{and} \tag{20}$$

$$(4)\ \text{Median}:\quad f_3(0) = 0. \tag{21}$$

The decision-maker's selection of a specific function satisfying the conditions presented in Eqs. (18)–(21) reflects his or her preferences. The transformation from LPR to DPR is defined using the abstract function described in Definition 10.

Definition 11. *Under the conditions described in Definition 10, with an abstract function satisfying the conditions presented in Eqs. (18)–(21), the LPR* $l^j_{i,lm}$ *is transformed into a DPR* $d^j_{i,lm} = \{(H_n, d^j_{n,i,lm}), n = 1, \ldots, N\}$, *with.*

$$d^j_{n,i,lm} = \begin{cases} \dfrac{f_3(\alpha) - s(H_{n-1})}{s(H_n) - s(H_{n-1})}, & s(H_{n-1}) \leqslant f_3(\alpha) < s(H_n) \\ \dfrac{s(H_{n+1}) - f_3(\alpha)}{s(H_{n+1}) - s(H_n)}, & s(H_n) \leqslant f_3(\alpha) < s(H_{n+1}). \\ 0, & \text{others} \end{cases} \tag{22}$$

Using Definition 11, the LPR matrices L^j_i ($i = 1, \ldots, L, j = q_3 + 1, \ldots, q_4$) can be transformed into the DPR matrices D^j_i. The internal consistency and restricted max-max transitivity of the transformation presented in Properties 2 and 3 are as follow:

Proposition 5. *Suppose, a DPR matrix* $D^j_i = (d^j_{i,lm})_{M\times M}$ *is transformed from an LPR matrix* $L^j_i = (l^j_{i,lm})_{M\times M}$ *using Definitions 10 and 11. Under these conditions, the score of* $d^j_{i,lm}$, $S^{j,L}_{i,lm}$ *was calculated using Eqs. (2)-(6). Then, the internal consistency between* L^j_i *and* D^j_i *is satisfied when.*

(1) $S^{j,L}_{i,lm} > S^{j,L}_{i,ml}$ if $l^j_{i,lm} > s_0$,
(2) $S^{j,L}_{i,lm} = S^{j,L}_{i,ml}$ if $l^j_{i,lm} = s_0$, and
(3) $S^{j,L}_{i,lm} < S^{j,L}_{i,ml}$ if $l^j_{i,lm} < s_0$.

Proposition 6. *Supposed that a DPR matrix* $D^j_i = (d^j_{i,lm})_{M\times M}$ *is transformed from an LPR matrix* $L^j_i = (l^j_{i,lm})_{M\times M}$ *by using Definitions 10 and 11. Assuming that the scores of* $d^j_{i,lm}$, $d^j_{i,mn}$ *and* $d^j_{i,ln}$, *i.e.,* $S^{j,L}_{i,lm}$, $S^{j,L}_{i,mn}$ *and* $S^{j,L}_{i,ln}$ *are calculated using Eqs. (2)-(6). Under the conditions, the restricted max-max transitivity of* D^j_i *is said to be satisfied when.*

(1) $S^{j,L}_{i,ln} \geq \max\{S^{j,L}_{i,lm}, S^{j,L}_{i,mn}\}$ if $l^j_{i,lm} \geq s_0$, $l^j_{i,mn} \geq s_0$ and $l^j_{i,ln} \geq \max\{l^j_{i,lm}, l^j_{i,mn}\}$, and.
(2) $S^{j,L}_{i,ln} \leq \min\{S^{j,L}_{i,lm}, S^{j,L}_{i,mn}\}$ if $l^j_{i,lm} \leq s_0$, $l^j_{i,mn} \leq s_0$ and $l^j_{i,ln} \leq \min\{l^j_{i,lm}, l^j_{i,mn}\}$.

Sections A. 5 and A. 6 of Appendix A prove Propositions 5 and 6, respectively, which verify the rationality of the transformation from LPR to DPR presented in Definitions 10 and 11.

4.4. Estimation of the reliability of each criterion

As analyzed by Fu et al. [26], the reliability of decision-makers is important for MCGDM. Following the idea of Fu et al. [26] to construct the reliabilities of decision-makers, group analysis and discussion (GAD) is conducted to help rectify misunderstandings and biases among decision-makers. Suppose, the unified DPRs of decision-makers before and after GAD are represented by $d^j_{i,lm(0)}$ ($j = 1, \ldots, T$) and $d^j_{i,lm(1)}$. To measure the reliability of decision-maker t_j after GAD, the similarity between $d^j_{i,lm(0)}$ and $d^k_{i,lm(1)}$ ($k \neq j$) is required. Inspired by the difference measure between belief distributions [40], a similarity measure between $d^j_{i,lm(0)}$ and $d^k_{i,lm(1)}$ ($k \neq j$) is constructed.

Definition 12. *Given the DPRs* $d^j_{i,lm(0)}$ *and* $d^k_{i,lm(1)}$, *their similarity is defined as.*

$$S^{jk}_{i,lm(0)(1)} = 1 - \sqrt{\frac{\sum_{n=1}^{N-1}\sum_{r=n+1}^{N} \Delta d^{jk}_{n,i,lm(0)(1)} \Delta d^{jk}_{r,i,lm(0)(1)} (s(H_r) - s(H_n)) + \Delta d^{jk}_{\Omega,i,lm(0)(1)}}{1.125}}, \tag{23}$$

where $\Delta d^{jk}_{n,i,lm(0)(1)} = \left(d^j_{n,i,lm(0)} - d^k_{n,i,lm(1)}\right)^2$ ($n = 1, \ldots, N$) and $\Delta d^{jk}_{\Omega,i,lm(0)(1)} = \left(d^j_{\Omega,i,lm(0)} - d^k_{\Omega,i,lm(1)}\right)^2$.

A numerical example is presented to illustrate the similarity measure in Eq. (23).

Example 2. *Suppose,* $\Omega = \{H_1, H_2, \ldots, H_9\}$ *and* $s(H_n)$ ($n = 1, \ldots, 9$) = (-1, −0.75, −0.5, −0.25, 0, 0.25, 0.5, 0.75, 1). *Given two DPRs* $d^j_{i,lm(0)}$ *and* $d^k_{i,lm(1)}$ *on* Ω *such that.*

$d^j_{i,lm(0)} = \{(H_4, 0.2), (H_6, 0.2), (H_7, 0.4), (\Omega, 0.2)\}$ and.

$d^k_{i,lm(1)} = \{(H_6, 0.4), (H_7, 0.2), (H_9, 0.1), (\Omega, 0.3)\}$, the process of calculating the similarity between the two DPRs is presented in the following. According to Eq. (23), we have.

$\Delta d^{jk}_{4,i,lm(0)(1)}\Delta d^{jk}_{6,i,lm(0)(1)}(s(H_6) - s(H_4))$= (0.2–0)^2·(0.2–0.4)^2·(0.25 – (-0.25)) = 0.0008,

$\Delta d^{jk}_{4,i,lm(0)(1)}\Delta d^{jk}_{7,i,lm(0)(1)}(s(H_7) - s(H_4))$= (0.2–0)^2·(0.4–0.2)^2·(0.5 – (-0.25)) = 0.0012,

$\Delta d^{jk}_{4,i,lm(0)(1)}\Delta d^{jk}_{9,i,lm(0)(1)}(s(H_9) - s(H_4))$= (0.2–0)^2·(0–0.1)^2·(1 – (-0.25)) = 0.0005,

$\Delta d^{jk}_{6,i,lm(0)(1)}\Delta d^{jk}_{7,i,lm(0)(1)}(s(H_7) - s(H_6))$= (0.2–0.4)^2·(0.4–0.2)^2·(0.5–0.25) = 0.0004,

$\Delta d^{jk}_{6,i,lm(0)(1)}\Delta d^{jk}_{9,i,lm(0)(1)}(s(H_9) - s(H_6))$= (0.2–0.4)^2·(0–0.1)^2·(1–0.25) = 0.0003,

$\Delta d^{jk}_{7,i,lm(0)(1)}\Delta d^{jk}_{9,i,lm(0)(1)}(s(H_9) - s(H_7))$= (0.4–0.2)^2·(0–0.1)^2·(1–0.5) = 0.0002, and.

$\Delta d^{jk}_{\Omega,i,lm(0)(1)}$= (0.2–0.3)^2 = 0.01.

As a whole, we have $S^{jk}_{i,lm(0)(1)}$ = 1 – ((0.0008 + 0.0012 + 0.0005 + 0.0004 + 0.0003 + 0.0002 + 0.01) / 1.125)$^{1/2}$ = 0.8909.

Example 2 shows the detailed calculation process for the similarity between two DPRs. It is satisfied that $0 \le S^{jk}_{i,lm(0)(1)} \le 1$, $S^{jk}_{i,lm(0)(1)} = 1$ when $d^j_{i,lm} = d^k_{i,lm}$, and $S^{jk}_{i,lm(0)(1)} = 0$ when $d^j_{i,lm} = \{(H_1, 0.5), (H_N, 0.5)\}$ (or $\{(\Omega, 1)\}$) and $d^k_{i,lm} = \{(\Omega, 1)\}$ (or $\{(H_1, 0.5), (H_N, 0.5)\}$). Using $S^{jk}_{i,lm(0)(1)}$, the reliability of decision-maker t_j after GAD is calculated by (Fu et al., 2015)

$$R^j_{i,lm} = \frac{\sum_{k=1,k\neq j}^{T} S^{jk}_{i,lm(0)(1)} \cdot R^k_{i,lm(0)}}{T-1} \tag{24}$$

with

$$R^k_{i,lm(0)} = \frac{\sum_{j=1,j\neq k}^{T} S^{kj}_{i,lm(0)(0)}}{T-1}. \tag{25}$$

Using the evidential reasoning (ER) rule [41] with $\lambda(e_i)$ and $R^j_{i,lm}$, $d^j_{i,lm(1)}$ ($j = 1, \ldots, T$) is combined to generate the group DPR $d_{i,lm} = \{(H_n, d_{n,i,lm}), n = 1, \ldots, N; (\Omega, d_{\Omega,i,lm})\}$ and the group reliability $R^{E(T)}_{i,lm}$. To further combine $d_{i,lm}$, using the ER rule with w_i and $R^{E(T)}_{i,lm} \cdot r_{i,lm}$, the reliability of criterion e_i, $r_{i,lm}$ needs to be determined. This represents the degree to which $d_{i,lm}$ determines the aggregated DPR d_{lm}. Without d_{lm}, it is difficult to directly estimate $r_{i,lm}$. An alternative choice is to estimate $r_{i,lm}$ for decision-maker t_j, $r^j_{i,lm}$, and further derive $r_{i,lm}$ from $r^j_{i,lm}$.

Definition 13. *The reliability of a criterion associated with a decision-maker and a pair of alternatives is estimated by combining of the similarity between the decision-maker's assessment and the assessment of any other decision-maker after the GAD.*

The estimation of $r^j_{i,lm}$ using Definition 13 is beneficial for generating group-satisfactory solutions. When $d^j_{i,lm(1)}$ is significantly different from $d^k_{i,lm(1)}$ ($k \neq j$), and $d^j_{i,lm(1)}$ greatly contributes to d_{lm}, the solution generated may not be satisfactory to decision-makers t_k ($k \neq j$). In contrary, when $d^j_{i,lm(1)}$ is close to $d^k_{i,lm(1)}$ ($k \neq j$) and $d^j_{i,lm(1)}$ greatly contributes to d_{lm}, the solution generated may be satisfactory to all decision-makers. According to Definition 13, the $r^j_{i,lm}$ is estimated as follows:

$$r^j_{i,lm} = \frac{\sum_{k=1,k\neq j}^{T} S^{jk}_{i,lm(1)(1)}}{T-1}, \tag{26}$$

where $S^{jk}_{i,lm(1)(1)}$ represents the similarity between DPRs $d^j_{i,lm(1)}$ and $d^k_{i,lm(1)}$, calculated using Definition 12.

Clearly, that $0 \le r^j_{i,lm(1)} \le 1$. In theory, $\frac{\sum_{j=1}^{T} r^j_{i,lm}}{T}$ can be considered as an estimation of r_i. However, the divergence among $r^j_{i,lm}$ was not considered. Centralized $r^j_{i,lm}$ ($j = 1, \ldots, T$) is more beneficial than divergent $r^j_{i,lm}$ ($j = 1, \ldots, T$) for generating group-satisfactory solutions. Considering this fact, $r_{i,lm}$ is estimated based on $r^j_{i,lm}$ using

$$r_{i,lm} = \left(\frac{\sum_{j=1}^{T} r^j_{i,lm}}{T}\right) \cdot \left(1 - \frac{\sqrt{\sum_{j=1}^{T}\left(r^j_{i,lm} - \sum_{k=1}^{T} r^k_{i,lm}\Big/T\right)^2 \Big/ (T-1)}}{\sigma_{\max}}\right), \tag{27}$$

where $\sigma_{\max}$ represents the maximum value of $\sqrt{\sum_{j=1}^{T}\left(r^j_{i,lm} - \sum_{k=1}^{T} r^k_{i,lm}/T\right)^2/(T-1)}$.

Proposition 7. *Given $\sigma = \sqrt{\sum_{j=1}^{T}\left(r^j_{i,lm} - \sum_{k=1}^{T} r^k_{i,lm}/T\right)^2/(T-1)}$, its maximum value is reached when $r^j_{i,lm} = 1$ ($j = 1, \ldots, (T+1)/2$) and $r^j_{i,lm} = 0$ ($j = (T+1)/2+1, \ldots, T$) or $r^j_{i,lm} = 1$ ($j = 1, \ldots$, (T-1)/2) and $r^j_{i,lm} = 0$ ($j =$ (T-1)/2 + 1, …, T) with T being odd, or $r^j_{i,lm} = 1$ ($j = 1, \ldots$, T/2) and $r^j_{i,lm} = 0$ ($j = T/2 + 1, \ldots, T$) with T being even.*

This proposition is proven in Section A.7 of Appendix A.

4.5. Process of the proposed method

The proposed method is based on the transformation of MPRs, IFPRs, and LPRs into DPRs and the estimation of criterion reliabilities, as shown in Fig. 1.

The proposed method includes three hierarchies: the basis, unification, and decision hierarchies. In the basis hierarchy, M alternatives x_l ($l = 1, \ldots, M$), L criteria e_i ($i = 1, \ldots, L$), T decision makes t_j ($j = 1, \ldots, T$) and a facilitator are involved in the MCGDM. The criterion weights w_i ($i = 1, \ldots, L$) and scores of grades $s(H_n)$ ($n = 1, \ldots, N$) are determined by the facilitator and the decision-makers, whereas the decision-makers' weights $\lambda^j(e_i)$ ($j = 1, \ldots, T$, $i = 1, \ldots, L$) are determined by the facilitator. Any of the five types of preference relations can be adopted by the decision-makers to express their preference information for each pair of alternatives.

In the unification hierarchy, before and after GAD, decision-makers give two rounds of preference relations, according to their preferred styles. The decision-makers preferring FPRs, IFPRs, and LPRs determine three specific functions that satisfy the conditions presented in Definitions 6, 8, and 10. The FPRs, IFPRs, and LPRs of the decision-makers are unified as DPRs using Definitions 7, 9, and 11, respectively, with their corresponding specific functions. The internal consistency and restricted max-max transitivity are ensured by the unification.

In the decision hierarchy, the unified DPRs of the decision-makers, before and after GAD, are used to calculate their reliabilities using Eqs. (23)-(25). The unified DPRs of the decision-makers after GAD are combined using the ER rule with their weights and reliabilities to generate group DPRs and group reliabilities on each criterion. The criterion reliabilities are estimated using Definitions 12–13 and Eqs. (26)-(27). Group DPRs are combined using the ER rule with criterion weights and reliabilities, group reliabilities to generate aggregated DPRs. Based on the score intervals of the aggregated DPRs, the complete and consistent score intervals of all pairs of alternatives are obtained using the method described by Fu et al. [9]. The scores of all the pairs of alternatives are then derived from

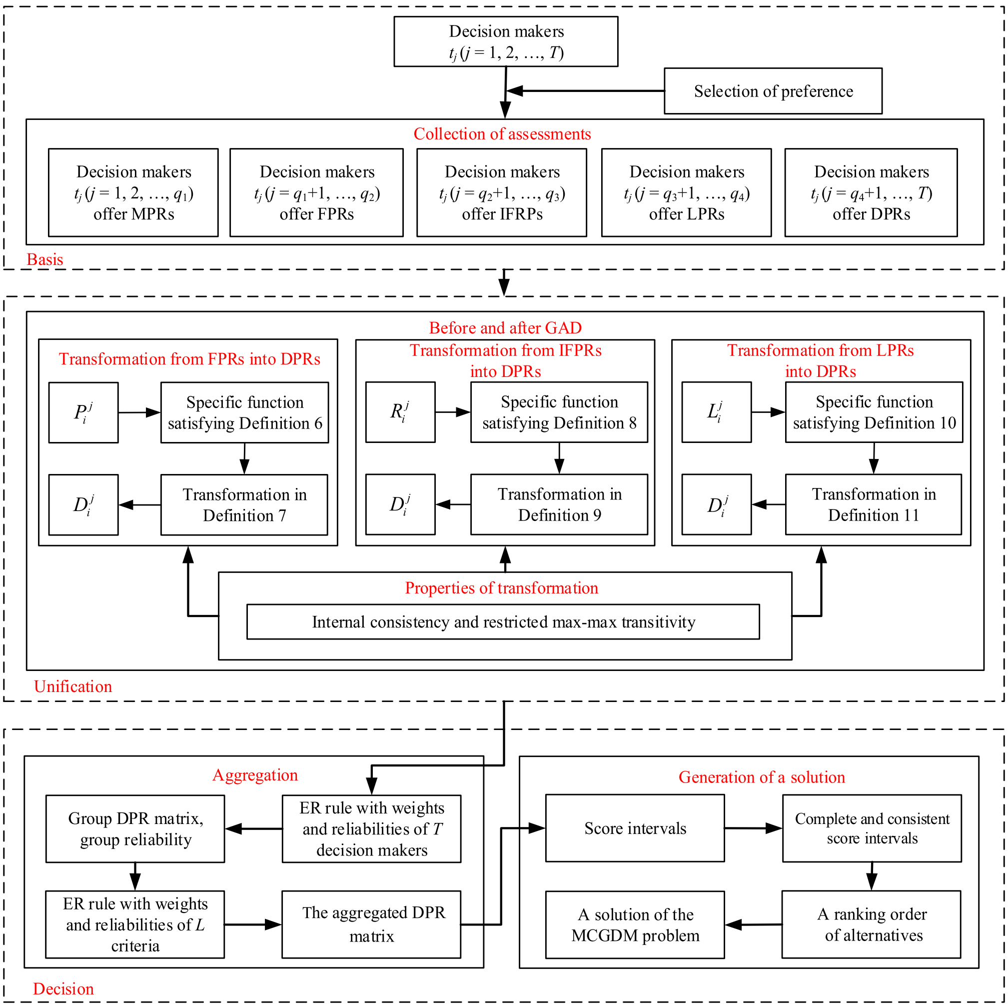

Fig. 1. Process of the proposed method.

their score intervals using Eq. (6), from which the ranking order of alternatives is generated. A final solution to the MCGDM problem is generated from the ranking order of the alternatives. These three hierarchies are summarized in Algorithm 1.

Algorithm 1: Process of the proposed model

Input: $A_i^j = (a_{i,lm}^j)_{M\times M}$ $(j = 1, \ldots, q_1)$ (MPR matrix);
$P_i^j = (p_{i,lm}^j)_{M\times M}$ $(j = q_1 + 1, \ldots, q_2)$ (FPR matrix);
$R_i^j = (r_{i,lm}^j)_{M\times M}$ $(j = q_2 + 1, \ldots, q_3)$ (IFPR matrix);
$L_i^j = (l_{i,lm}^j)_{M\times M}$ $(j = q_3 + 1, \ldots, q_4)$ (LPR matrix);
$D_i^j = (d_{i,lm}^j)_{M\times M}$ $(j = q_4 + 1, \ldots, T)$ (DPR matrix).
Output: The ranking order of the M alternatives.
Process:
1: Determine w_i $(i = 1, \ldots, L)$, $s(H_n)$ $(n = 1, \ldots, N)$, and $\lambda^j(e_i)$ $(j = 1, \ldots, T, i = 1, \ldots, L)$;
2: Determine $f_1(x)$, $f_2(x, y)$, and $f_3(x)$ satisfying the conditions presented in Definitions 6, 8, and 10, respectively;
3: Transform MPR, FPR, IFPR, and LPR into DPR using Definitions 7, 9, and 11 with $f_1(x)$, $f_2(x, y)$, and $f_3(x)$, respectively;
4: Use the unified DPRs of the decision-makers before and after GAD to obtain $R_{i,lm}^j$ $(j = 1, \ldots, T, i = 1, \ldots, L, l = 1, \ldots, M, m = 1, \ldots, M)$ by using Eqs. (23)-(25);
5: Generate group DPRs $d_{i,lm} = \{(H_n, d_{n,i,lm}), n = 1, \ldots, N; (\Omega, d_{\Omega,i,lm})\}$ and the group reliability $R_{i,lm}^{E(T)}$ with $\lambda(e_i)$ and $R_{i,lm}^j$;
6: Generate the aggregated DPRs with w_i and $R_{i,lm}^{E(T)} \cdot r_{i,lm}$ by using Eqs. (26)-(27);
7: Generate the score intervals of the aggregated DPRs by using Eqs. (2)-(3);
8: Generate the scores of all pairs of alternatives by using Eq. (6);
9: Use the scores of all pairs of alternatives to generate the ranking order of the M alternatives.

Finally, the computational complexity of the indirect MCGDM method is analyzed from three perspectives. The first is the transformation of the preference relations. The relevant computational complexity is calculated as $O_1 = O((T - q_4) \cdot (M - 1) \cdot L)$. The second perspective involves the calculation of criterion reliability. The relevant computational complexity is calculated as $O_2 = O(T \cdot (T - 1) \cdot (M - 1) \cdot L + (T + 1) \cdot (M - 1) \cdot L + (L \cdot N + 2 \cdot (N + 2)) \cdot (M - 1) \cdot L)$. The third perspective involves the generation of an aggregated DPR. The relevant computational complexity is calculated as $O_3 = O((L \cdot N + 2 \cdot (N + 2)) \cdot (M - 1))$. Note that O_1 and O_3 are significantly smaller than O_2; thus, the computational complexity of the indirect MCGDM method is approximate to O_2.

5. Case study

In this section, the problem of selecting a new business to establish a unitary entrepreneurial organization for an enterprise located in Changzhou, Jiangsu, China is solved using the proposed method to demonstrate its applicability.

5.1. Description of the problem of selecting a new business

A unitary entrepreneurial organization is beneficial for the innovative and sustainable development of an enterprise, which is consistent with its strategic objectives. Such an organization depends on the construction of a new business. However, constructing a new business profoundly depends on investments in personnel, capital, brand image, and other enterprise resources, which may lead to investment risk. To effectively avoid and control investment risk, a new business should be deliberately selected to establish a unitary entrepreneurial organization.

An enterprise located in Changzhou, Jiangsu, China pursues innovative and suitable development. For this purpose, the enterprise plans to establish a unitary entrepreneurial organization by establishing a new business. Considering the familiarity with the resources related to new businesses, the deputy general manager of the enterprise responsible for management innovation acts as the facilitator. He has identified five new businesses, whose construction may lead to the establishment of a unitary entrepreneurial organization. The five new businesses are the RV reducer of an industrial robot (B_1), the reducer of a shield-driven machine (B_2), a cleaning machine with high pressure water jet (B_3), an intelligent parking lock (B_4), and a driving system of new energy automobile (B_5).

The facilitator invited the headers of the eight departments as decision-makers to deliberately select a new business from the five. These are the header of human resource department (t_1), the header of manufacturing and process department (t_2), the header of planning and development department (t_3), the header of operation management department (t_4), the header of science and technology management department (t_5), the header of technology R&D center (t_6), the header of new industry development department (t_7), and the header of finance department (t_8). Five new businesses were evaluated based on the seven criteria presented in Table 1. The method of Ölçer and Odabasi [42] was used by the facilitator to determine the relative weights of the eight decision-makers on the seven criteria presented in Table 2.

The eight decision-makers were encouraged to freely choose one of MPR, FPR, IFPR, LPR, or DPR to evaluate the five new businesses. Decision-makers t_1 and t_2 selected MPRs; decision-makers t_3 and t_4 selected FPRs; decision-makers t_5 and t_6 selected IFPRs; decision-maker t_7 selected LPRs; and decision-maker t_8 selected DPRs. Decision-maker t_7 evaluates the five new businesses, using the subscript-symmetric additive linguistic term set $S = \{s_{-3}, s_{-2}, s_{-1}, s_0, s_1, s_2, s_3\}$ = {very poor, poor, slightly poor, fair, slightly good, good, very good}. Decision-maker t_8 evaluated the five new businesses, using the set of grades $\Omega = \{H_1, H_2, \ldots, H_9\}$ = {absolutely weaker, much weaker, moderately weaker, marginally weaker, indifferent, marginally stronger, moderately stronger, much stronger, absolutely stronger}.

Following a discussion among the facilitator and the eight decision-makers regarding the seven criteria, criterion weights were specified as $w = (0.09, 0.11, 0.19, 0.12, 0.18, 0.14, 0.17)$, using the method of Ölçer and Odabas [42], and the scores of grades H_n ($n = 1, \ldots, 9$) were set as $s(H_n) = (-1, -0.75, -0.5, -0.25, 0, 0.25, 0.5, 0.75, 1)$, using the probability assignment approach [43]. Using the five types of preference relations, the eight decision-makers offered the first round of preference relations between adjacent new businesses based on the seven criteria, as shown in Tables B.1-B.8 of Section B.1 in Appendix B of the Supplementary Material. They also provided renewed preference relations between adjacent new businesses based on the seven criteria to determine criterion reliability, as shown in Tables B.9-B.16 of Section B.1.

5.2. Generation of a solution to the problem of selecting a new business

The eight decision-makers needed to choose specific functions to unify their preference information, by using the proposed method of analysis. The function $f_1(x) = 2x–1$ satisfying the conditions presented in Definition 6 was adopted by decision-makers t_j ($j = 1, \ldots, 4$); the function $f_2(x, y) = \frac{x-y}{x+y}$ satisfying the conditions presented in Definition 8 was selected by decision-makers t_j ($j = 5, 6$); and the function $f_3(x) = \frac{x}{3}$ satisfying the conditions presented in Definition 10 was selected by decision-maker t_7.

The MPR matrices of decision-makers t_1 and t_2 presented in Tables B.1-B.2 and B.9-B.10 are transformed into DPR matrices using the function $f_1(x)$ and Definitions 2 and 7, as presented in Tables B.17-B.20 of Section B.2 in Appendix B of the Supplementary Material. The FPR matrices of decision-makers t_3 and t_4 presented in Tables B.3-B.4 and B.11-B.12 are transformed into DPR matrices using the function $f_1(x)$ and Definition 7, as presented in Tables B.21-B.24 of Section B.2. The IFPR matrices of decision-makers t_5 and t_6 presented in Tables B.5-B.6 and B.13-B.14 are transformed into DPR matrices using the function $f_2(x, y)$ and Definition 9, as presented in Tables B.25-B.28 of Section B.2. The LPR matrices of decision-maker t_7 presented in Table B.7 and B.15 are transformed into DPR matrices using the function $f_3(x)$ and Definition 11, as presented in Tables B.29-B.30 of Section B.2.

The unified DPR matrices of the eight decision-makers, the reliability of decision-maker $R^j_{i,l(l+1)}$ ($j = 1, \ldots, 8$, $i = 1, \ldots, 7$, $l = 1, \ldots, 4$) was calculated using Eqs. (23)-(25). The unified DPRs $d^j_{i,l(l+1)(1)}$ ($j = 1, \ldots, 8$, $i = 1, \ldots, 7$, $l = 1, \ldots, 4$) were combined using the ER rule [41] with $\lambda(e_i)$ ($i = 1, \ldots, 7$) and $R^j_{i,l(l+1)}$ to generate the group DPR $d_{i,l(l+1)} = \{(H_n, d_{n,i,l(l+1)}), n = 1, \ldots, 9; (\Omega, d_{\Omega,i,l(l+1)})\}$ and group reliability $R^{E(8)}_{i,l(l+1)}$. The criterion reliability $r_{i,l(l+1)}$ ($i = 1, \ldots, 7$, $l = 1, \ldots, 4$) was calculated using Eqs. (26)-(27). Subsequently, the group DPR $d_{i,l(l+1)}$ ($i = 1, \ldots, 7$, $l = 1, \ldots, 4$) was combined using the ER rule [41] with w_i ($i = 1, \ldots, 7$) and $R^{E(8)}_{i,l(l+1)} \cdot r_{i,l(l+1)}$ to generate the aggregated DPR $d_{l(l+1)}$ ($l = 1, \ldots, 4$), as presented in Table B.31 of Section B.2.

The score intervals of adjacent new businesses $s_{l(l+1)}$ ($l = 1, \ldots, 4$) were obtained from the aggregated DPR $d_{l(l+1)}$ using Eqs. (2)-(3). The consistent score intervals of non-adjacent new businesses s_{lm} ($|l\text{-}m| > 1$) ($l, m = 1, \ldots, 5$) were obtained from $s_{l(l+1)}$, using the method proposed by Fu et al. [9]. The detailed process was described in Section B.3 of Appendix B. The consistent score intervals for any pair of eight new businesses were presented in Table 3. The scores for each new business over others were then obtained using Eq. (6) from the score intervals presented in Table 3, as listed in Table 4. The ranking order of the five new businesses was generated from the scores presented in Table 4, which was $B_5 \succ B_1 \succ B_4 \succ B_2 \succ B_3$ with "$\succ$" denoting "superior to". The driving system of a new energy automobile (B_5) was selected by eight decision-makers to establish a unitary entrepreneurial organization.

5.3. Influence of different unification functions on the solution

As presented in Section 5.2, three specific functions were selected by the first seven decision-makers to transform FPRs, IFPRs, and LPRs to DPRs. An interesting issue is whether different specific functions could result in different ranking orders for the five new businesses. To address this issue, several possible choices for $f_1(x)$, $f_2(x, y)$, and $f_3(x)$ were presented in Table 5. Based on the possible unification functions presented in Table 5, eight combinations of $f_1(x)$, $f_2(x, y)$, and $f_3(x)$ were selected to generate the ranking order of the five new businesses. The results are summarized in Table 6.

Table 6 shows that five different ranking orders of the five new businesses are generated from the eight combinations of $f_1(x)$, $f_2(x, y)$, and $f_3(x)$. B_1 is the best business under the 5th combination of $f_1(x)$, $f_2(x, y)$, and $f_3(x)$, whereas B_5 becomes the best business under the other seven combinations. While B_3 is the worst business under the 1st, 3rd, 4th, and 5th combinations of $f_1(x)$, $f_2(x, y)$, and $f_3(x)$, and B_2 becomes the worst business under the 2nd, 6th, and 7th combinations. Particularly, B_1 changes to be the worst business under the 8th combination of $f_1(x)$, $f_2(x, y)$, and $f_3(x)$. These results highlight the important influence of specific functions of decision-makers on the rankings of new businesses.

Table 1
Description of the seven criteria.

Criteria	Description
e_1	Ability to conform to the policies of the nation and the industry
e_2	Ability to meet the development strategy of the enterprise
e_3	Future growth of new business
e_4	Overall profitability of the new business
e_5	Degree of commonality in basic technology and applied technology
e_6	Degree of commonality in marketing resources and industry resources
e_7	Degree of loss caused by the implementation of the new business

Table 2
Relative weights of the eight decision-makers.

Criteria	$\lambda^1(e_i)$	$\lambda^2(e_i)$	$\lambda^3(e_i)$	$\lambda^4(e_i)$	$\lambda^5(e_i)$	$\lambda^6(e_i)$	$\lambda^7(e_i)$	$\lambda^8(e_i)$
e_1	0.12	0.14	0.13	0.08	0.11	0.18	0.15	0.09
e_2	0.11	0.09	0.17	0.13	0.15	0.12	0.16	0.07
e_3	0.07	0.11	0.22	0.15	0.14	0.11	0.13	0.07
e_4	0.12	0.08	0.16	0.13	0.11	0.09	0.12	0.19
e_5	0.09	0.08	0.12	0.17	0.13	0.14	0.11	0.16
e_6	0.18	0.13	0.14	0.09	0.1	0.17	0.11	0.08
e_7	0.12	0.16	0.11	0.07	0.16	0.15	0.14	0.09

Table 3
Score intervals of any pairs of new businesses.

Businesses	B_1	B_2	B_3	B_4	B_5
B_1	–	[0.0768, 0.1597]	[0.0606, 0.2194]	[-0.0912, 0.1591]	[-0.2239, 0.0190]
B_2	[-0.1597, −0.0768]	–	[-0.0162, 0.0657]	[-0.1667, 0.0055]	[-0.2923, −0.0290]
B_3	[-0.2194, −0.0606]	[-0.0657, 0.0162]	–	[-0.1518, −0.0603]	[-0.2789, −0.0937]
B_4	[-0.1591, 0.0912]	[-0.0055, 0.1667]	[0.0603, 0.1518]	–	[-0.1401, −0.0345]
B_5	[-0.0190, 0.2239]	[0.0290, 0.2923]	[0.0937, 0.2789]	[0.0345, 0.1401]	–

Table 4
Scores of any pairs of new businesses.

Businesses	B_1	B_2	B_3	B_4	B_5
B_1	0	0.1134	0.1289	0.0297	−0.0900
B_2	−0.1134	0	0.0238	−0.0736	−0.1395
B_3	−0.1289	−0.0238	0	−0.1012	−0.1690
B_4	−0.0297	0.0736	0.1012	0	−0.0827
B_5	0.0900	0.1395	0.1690	0.0827	0

5.4. Sensitivity analysis of group reliability and criterion reliability

As indicated in Section 4.5, the group reliability and criterion reliability contribute to the generation of a final solution for the proposed method. Sensitivity analyses were conducted to examine the contributions of these factors.

The first focus was on the sensitivity analysis of group reliability. Let us assume that group reliability $R^{E(8)}_{7,l(l+1)}$ ($l = 1, \ldots, 4$) changed from 0 to 1 with a step of 0.1. Under these conditions, the rankings of the five new businesses were recalculated and plotted in Fig. 2. As shown in Fig. 2, B1 remained the best business when $R^{E(8)}_{7,l(l+1)}$ increased from 0 to 0.7, and became the 3rd best business when $R^{E(8)}_{7,l(l+1)}$ increased to 0.8. The rankings of B_4 and B_5 changed when $R^{E(8)}_{7,l(l+1)}$ increased to 0.8 and those of B_2 and B_3 changed when $R^{E(8)}_{7,l(l+1)}$ increased to 0.9. These findings highlight the contribution of group reliability to the rankings of the five new businesses.

The second focus was on the sensitivity analysis of criterion reliability. Criterion reliability $r_{5,l(l+1)}$ ($l = 1, \ldots, 4$) was assumed to change from 0 to 1 with a step of 0.1. The rankings of the five new businesses were recalculated and plotted in Fig. 3. It shows that B_5 remains to be the best business when $r_{5,l(l+1)} \leq 0.9$ and changes to be the 2nd best business when $r_{5,l(l+1)} > 0.9$. The rankings of the other four new businesses changed when $r_{5,l(l+1)}$ increased to 0.3. The changes in the rankings of the five new businesses with the variation in $r_{5,l(l+1)}$ highlight the contribution of criterion reliability to the ranking order of the five new businesses.

Table 5
Possible choices of three types of unification functions.

Unification functions	Possible choices
$f_1^k(x)(k = 1, \ldots, 4)$	$2x–1$, $(2x–1)^{1/3}$, $\sin\left(\pi x - \frac{\pi}{2}\right)$,$\tan\left(\frac{\pi}{2}x - \frac{\pi}{4}\right)$
$f_2^k(x,y)(k = 1, \ldots, 4)$	$\frac{x-y}{x+y}$, $\left(\frac{x-y}{x+y}\right)^{1/3}$, $\sin\left(\frac{\pi(x-y)}{2(1-\pi)}\right)$,$\tan\left(\frac{\pi(x-y)}{4(1-\pi)}\right)$
$f_3^k(x)(k = 1, \ldots, 3)$	$\frac{x}{3}$, $\sin\left(\frac{\pi}{6}x\right)$,$\tan\left(\frac{\pi}{12}x\right)$

Table 6
Ranking orders of the five new businesses based on the eight combinations of $f_1(x)$, $f_2(x, y)$, and $f_3(x)$.

No	Combination of $f_1(x)$, $f_2(x, y)$, and $f_3(x)$	Ranking order of the five new businesses
1	$f_1^1(x), f_2^1(x,y), f_3^1(x)$	$B_5 \succ B_1 \succ B_4 \succ B_2 \succ B_3$
2	$f_1^1(x), f_2^4(x,y), f_3^1(x)$	$B_5 \succ B_4 \succ B_3 \succ B_1 \succ B_2$
3	$f_1^2(x), f_2^1(x,y), f_3^1(x)$	$B_5 \succ B_1 \succ B_2 \succ B_4 \succ B_3$
4	$f_1^2(x), f_2^1(x,y), f_3^2(x)$	$B_5 \succ B_1 \succ B_2 \succ B_4 \succ B_3$
5	$f_1^2(x), f_2^2(x,y), f_3^3(x)$	$B_1 \succ B_5 \succ B_4 \succ B_2 \succ B_3$
6	$f_1^2(x), f_2^3(x,y), f_3^1(x)$	$B_5 \succ B_4 \succ B_3 \succ B_1 \succ B_2$
7	$f_1^3(x), f_2^4(x,y), f_3^1(x)$	$B_5 \succ B_4 \succ B_3 \succ B_1 \succ B_2$
8	$f_1^4(x), f_2^3(x,y), f_3^2(x)$	$B_5 \succ B_4 \succ B_3 \succ B_2 \succ B_1$

5.5. *Comparison with existing methods*

To demonstrate the advantages of the proposed method, it is compared with existing MCGDM methods in [29–30,44–46], qualitatively and quantitatively. A qualitative comparison was conducted from the perspective of internal consistency and restricted max-max transitivity. The details are listed in Table 7.

Table 7 indicates that only the proposed method can simultaneously satisfy internal consistency and restricted max-max transitivity. The five existing methods do not satisfy restricted max-max transitivity. However, restricted max-max transitivity is stricter than weak transitivity and more consistent with the intuitive sense of decision-makers [35]. This is a necessary condition for satisfying the consistency of preference relations [47]. If restricted max-max transitivity is not satisfied, inconsistent preference relations may be generated [48] to lead to unreasonable solutions.

As shown in Table 7, there are two or three identical types of preference relations in the methods in [30,44–45] and the proposed method. To facilitate quantitative comparison, two or three of MPR, FPR, and LPR are selected to characterize decision-makers' preferences in the methods in [30,44–45] and the proposed method. In this situation, the ranking orders of the five new businesses are generated using the methods in [30,44–45], as presented in Table 8. Under the condition that two or three of MPR, FPR, and LPR are unified as DPRs based on the settings specified in Section 5.1, the ranking orders of the five new businesses are regenerated using the proposed method and are presented in Table 8.

Table 8 shows the clear differences between the three existing MCGDM methods and the proposed method. In a situation where MPR and FPR are selected to characterize the preferences of decision-makers, the best business determined by the proposed method is different from that derived from the methods in [30,45]. There were also significant differences between the ranking orders of the five businesses generated per the three methods. When LPR was added to characterize the preferences of decision-makers, both the best business and the ranking order of the five businesses generated by the proposed method were significantly different from those derived from the method in [44]. These differences originate from two key points of this study: first, the consideration of decision-makers' preferences in the unification of MPR, FPR, and LPR, second, the consideration of decision-makers' reliabilities in the aggregation of preference information.

Overall, qualitative and quantitative comparisons of the proposed method with existing MCGDM methods indicate that the proposed method is beneficial for generating reasonable solutions by guaranteeing consistent preference relations, involving decision-makers' preferences in the unification of heterogeneous preference relations, and considering decision-makers' reliabilities in the generation of aggregated preference information.

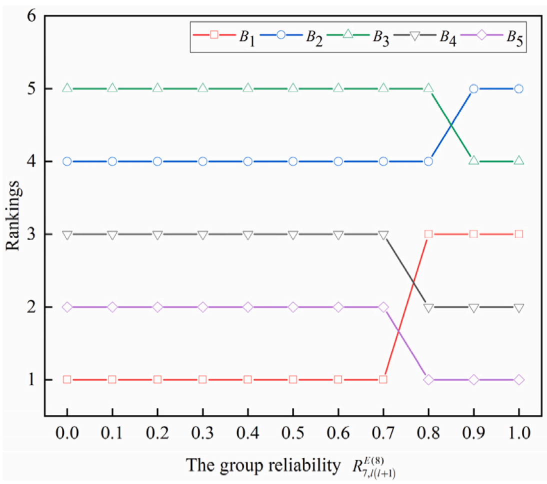

Fig. 2. Movement of the rankings of the five new businesses with the variation in $R^{E(8)}_{i,l(l+1)}$.

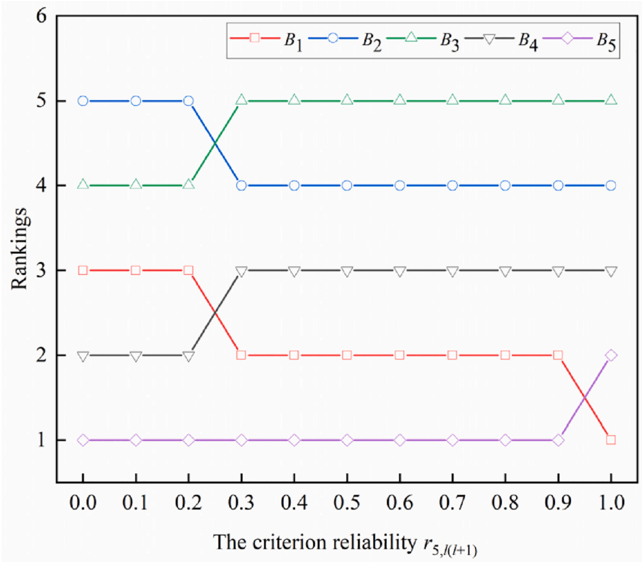

Fig. 3. Movement of the rankings of the five new businesses with the variation in $r_{i,l(l+1)}$.

Table 7
Qualitative comparison of the proposed method with existing MCGDM methods.

Existing methods	Heterogeneous preference relations	Internal consistency	Restricted max-max transitivity
Method in [29]	Preference orderings, utility functions, MPRs	Yes	No
Method in [30]	Preference orderings, utility functions, MPRs, FPRs	Yes	No
Method in [44]	MPRs, FPRs, LPRs	Yes	No
Method in [45]	MPRs, FPRs, additive LPRs, multiplicative LPRs	Yes	No
Method in [46]	IFPRs, multiplicative IFPRs, additive linguistic IFPRs, multiplicative linguistic IFPRs	Yes	No
The proposed method	MPRs, FPRs, IFPRs, LPRs, DPRs	Yes	Yes

6. Conclusions

This study proposes a new indirect method to address MCGDM problems with five different types of preference relations and decision-makers' reliabilities. The reason behind the unification of the five types of preference relations has qualitatively been analyzed. Based on this analysis, the transformations from MPRs, FPRs, IFPRs, and LPRs to DPRs are theoretically constructed and verified to satisfy two important properties: internal consistency and restricted max-max transitivity. Meanwhile, the reliability of each criterion has, reasonably and efficiently, been estimated to ensure the rational involvement of decision-makers' reliabilities in MCGDM.

The contributions of this study include the following: (1) an indirect method is proposed to solve MCGDM problems with heterogeneous preference relations and decision-makers' reliabilities; (2) an abstract transformation from MPRs, FPRs, IFPRs, and LPRs to DPRs is constructed, in which the internal consistency and restricted max-max transitivity are satisfied simultaneously; (3) criterion reliabilities are estimated based on the calculation of decision-makers' reliabilities; and (4) the proposed method is used to solve the problem of selecting a new business to establish a unitary entrepreneurial organization.

The limitations of this study are as follows: (1) parameters are not considered in the unification functions to support the transformation from MPRs, FPRs, IFPRs, and LPRs to DPRs; (2) tolerance thresholds and risk attitudes of decision-makers are not considered in the proposed method; and (3) other group decision-making elements, such as group consensus and group satisfaction are not considered in the proposed method.

Further, studies should be conducted to address these above limitations. In addition to theoretical innovations, various types of applications will be addressed to identify potential problems for analysis, which will stimulate theoretical studies.

CRediT authorship contribution statement

Chao Fu: Methodology, Software, Validation, Writing – original draft, Funding acquisition. **Xuefei Jia:** Methodology, Software, Writing – original draft. **Wenjun Chang:** Supervision, Writing – review & editing, Funding acquisition.

Table 8
Quantitative comparison of the proposed method with existing MCGDM methods.

Existing methods	Ranking order of the five new businesses
Method in [30] with MPRs and FPRs	$B_3 \succ B_2 \succ B_5 \succ B_4 \succ B_1$
Method in [45] with MPRs and FPRs	$B_1 \succ B_5 \succ B_4 \succ B_2 \succ B_3$
The proposed method with MPRs and FPRs	$B_5 \succ B_4 \succ B_1 \succ B_3 \succ B_2$
Method in [44] with MPRs, FPRs, and LPRs	$B_3 \succ B_2 \succ B_4 \succ B_5 \succ B_1$
The proposed method with MPRs, FPRs, and LPRs	$B_5 \succ B_4 \succ B_3 \succ B_1 \succ B_2$

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ins.2023.119492.

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